Coding Sample Output

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Overview

This report presents my analysis of the effect of Google's book digitization program on physical library loans at Harvard, based on a balanced panel of approximately 88,000 books over the 2003–2011 period.

- **Section 1** replicates the main results from the original study using log-OLS and linear probability models with book and year-location fixed effects.
- **Section 2** provides a robustness check using the Callaway & Sant'Anna (2021) estimator to address concerns about staggered treatment timing in two-way fixed effects.
- **Section 3** merges in book-category data from the study's replication package and examines heterogeneity in the digitization effect across genres, including a comparison bar chart and separate regressions for Art and Education books.

1 Replication of Table 5

From Table 1, it shows a replication of the main effects of digitization on Harvard library loans. Adding both the book-fixed effects and year-location fixed effects, the coefficients on post-scanned is about -0.0511 in the log-OLS model, meaning the digitization will decrease the log number of Harvard library loans by 5%. Similarly, the coefficient is -0.0613 in the LPM model, implying that digitization decreases the probability that a book is checked out at Harvard at all in a year by 6.1 percentage points. Both negative correlations are significant at the 0.01 level.

2 Robustness Check

To deal with the specific concern towards the weighting issues when using TWFE, I'm using the model from Callaway Sant'Anna (2021)[1]. This approach explicitly accounts for heterogeneous treatment effects across different timing groups and avoids the negative weighting problem of TWFE. It's implemented in Stata using the *csdid* package.

From Table 2, we can see that after we have captured the potential weighting issues generated from the two-way fixed effect model, using the model by Callaway and Sant'Anna, the

Table 1: Effects of Digitization on Harvard Library Loans

	(1) log-OLS	(2) LPM
post_scanned	-0.0511***	-0.0613***
	(0.00152)	(0.00170)
Book FE	Yes	Yes
Year-Location FE	Yes	Yes
Observations	792054	792054

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Average Treatment-on-the-Treated(ATT) coefficients stay approximately the same as our main result.

Therefore, our robustness check using the Callaway & Sant'Anna (2021) difference-in-differences estimator in Table 2 confirms the main findings by showing that the negative effect of digitization on physical book borrowing. The consistency in both the direction and approximate magnitude of the effects across different estimation approaches provides stronger evidence that digitization genuinely reduces physical book borrowing.

Table 2: Robustness check(using model from Callway(2021))

	ATT	Std. Error	p-value
Log-OLS(csdid)	-0.059***	0.002	0.000
LPM(csdid)	-0.070***	0.003	0.000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

3 Different effects across Categories

3.1 Top Categories

I merge the master dataset with the Wikipedia dataset that contains the book category. Table 3 represents the top category in this dataset. I will use some of the most common categories to do the analysis to make sure that the overall analysis is representative of the overall dataset. The categories I picked for further analysis are: History, Fiction, Science, Education, and Art.

3.2 Visualization of the Difference Across Different Type of Books

Figure 1 shows the comparison of average log loans across five book categories (Art, Education, Fiction, History, and Science) before and after digitization. The blue bars represent pre-digitization loans, while the red bars show post-digitization loans.

Table 3: Top 10 Book Categories by Frequency

Rank	Book Category	Count
1	History	7,245
2	United States	2,277
3	Fiction	2,007
4	Reference	1,827
5	Biography & Autobiography	1,611
6	Science	1,440
7	Law	1,359
8	Education	1,350
9	Art	1,197
10	Great Britain	1,170

We can see clearly from the graph: All categories show a decrease in loans after digitization. Among all, Art and Science show the largest decreases after digitization. Education shows the smallest decrease, maintaining the highest post-digitization loan level. In particular, we can see that the pre-digitization loan levels were fairly similar across categories (ranging from about 0.13 to 0.15), but the post-digitization levels vary much more (from about 0.06 for Art to 0.10 for Education).

Art and Education represent interesting contrasting cases for further regression analysis, as they show the largest and smallest effects of digitization, respectively. Art books experienced a dramatic drop in borrowing after digitization, while Education materials maintained relatively higher borrowing levels, suggesting different user responses to digitization across these disciplines. Therefore, I focused particularly on the comparison between art and education for the following regression analysis,

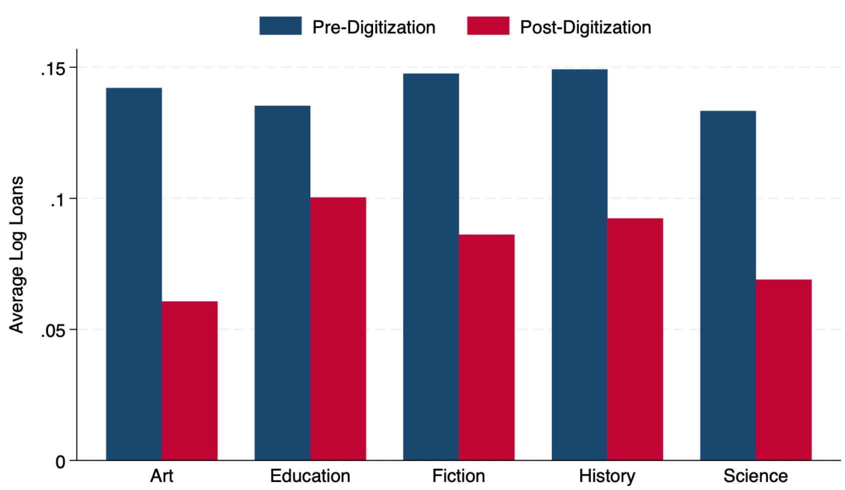

Figure 1: Average Book Loans Across Categories

3.3 Regression Analysis between Art and Education

Based on Table 4, the results show a clear difference in how digitization affected Art books versus Education books: For Art books (column 1), digitization led to a statistically significant decrease in log loans of 0.0924 ($p < 0.01$). This represents approximately a 9.2% reduction in borrowing after digitization, indicating a substantial negative impact. For Education books (column 2), digitization was associated with a smaller decrease in log loans of 0.0402, which is not statistically significant (no stars). This suggests that while there was a slight negative trend, we cannot confidently conclude that digitization affected Education book loans.

The contrasting results reveal that digitization had heterogeneous effects across book categories, with Art books experiencing a much stronger negative impact than Education books. This difference may reflect varying user preferences, the importance of physical interaction with Art materials, or different adaptation to digital formats across disciplines.

Table 4: Differential Effects of Digitization on Art vs. Education Books

	(1) Art	(2) Education
post_scanned	-0.0924***	-0.0402
	(0.0239)	(0.0264)
Book FE	Yes	Yes
Year-Location FE	Yes	Yes
N	504	558

Standard errors in parentheses

Book and Year-Location fixed effects included in all specifications.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

References

[1] Brantly Callaway and Pedro H.C. Sant'Anna. "Difference-in-Differences with multiple time periods". en. In: *Journal of Econometrics* 225.2 (Dec. 2021), pp. 200–230. ISSN: 03044076. DOI: 10.1016/j.jeconom.2020.12.001. URL: https://linkinghub.elsevier.com/retrieve/pii/S0304407620303948 (visited on 04/15/2025).